Vision Core v1.1

V1.1 designed by Cris R. Martinez Hernandez

```
=== NET LABEL NAMING CONVENTION ===
A/B suffix (ex. GPIO_14A, GPIO_14B):
Two pins that are physically related on the FPGA package (adjacent
pair number) but electrically INDEPENDENT, meaning each carries its own
unrelated signal. NOT a differential pair.

+/- suffix (ex. GPIO_8+, GPIO_8-):
A differential pair — the two pins are deliberately driven
as opposites of each other to carry ONE signal.

Whether a given physical pin pair is used as A/B or +/- is a design
choice made on the sheet where it's wired — not a fixed property of
the silicon. Check the destination, not just the pin name, if unsure.

Swapping + and - on a pair is allowed only if the far end is updated
to match (or the polarity is inverted in the FPGA logic).


=== VOLTAGE MAP ===
VOLTAGE MAP (VCCIO per bank):
Bank 0   3.3V          single-ended only — GPIO expansion header
Bank 1   3.3V          GPDI video output (emulated differential) + GPIO exp.
Bank 2   3.3V          SDRAM
Bank 3   3.3V          SDRAM
Bank 6   3.3V          USB-UART (FT231XS), SD card
Bank 7   2.5V / 3.3V   GPIO expansion header (+/- pairs), audio/video, LEDs
                       VCCIO7 = +2V5_3V3 (selectable rail). Verify the
                       selected voltage before connecting external signals.
Bank 8   3.3V          Dedicated config/JTAG pins (not general GPIO)

Other rails: VCC (core) 1.1V, VCCAUX 2.5V.
Signals may move between banks only if the banks share the same VCCIO.
Bank 7 signals at 2.5V are NOT 3.3V compatible.


=== PIN ASSIGNMENT METHODOLOGY ===
PIN ASSIGNMENT METHODOLOGY:
Source: Lattice ECP5 caBGA381 pin migration data (per-device column), this
spreadsheet confirms bank, dual function, and differential status for any
physical ball.

Refer to bank/voltage map first — confirms what's already committed before
hunting for a free pin.

Before assigning a pin, check:
  - bank: must match the voltage already committed to that bank
  - dual function: some balls double as clock inputs — don't collide
  - differential: only matters if building an actual diff pair (see
    A/B vs +/- note)

Before wiring any IC's pins, verify against its own datasheet, never assume
a reference design's values/pins carry over to a different part or voltage combo.

UNUSED GPIO PINS:
Unused ordinary GPIOs are left as No Connect — no external pull resistor
required. The ECP5 has internal weak pulls enabled by default on every I/O
(tri-stated with weak pull-down at power-up). Only add an external pull if
the internal one is intentionally disabled in the pin constraints file.
Source: Lattice Answer Database #738.

GPDI DIFFERENTIAL SIGNALING:
GPDI does NOT use the ECP5's dedicated True-LVDS hardware driver. It uses
"emulated" differential signaling — ordinary LVCMOS pins driven as a
complementary pair, using external resistors/caps to approximate the signal
levels. This works on any confirmed A/B pin pair, True-LVDS-flagged or not,
and needs no special 2.5V bank. Source: Lattice ECP5 sysI/O Usage Guide,
FPGA-TN-02032.

True LVDS output on the GPIO +/- pairs would require bank 7 at 2.5V AND a
pin pair that supports it. Confirm in the sysI/O Usage Guide before use.

=== PCB DESIGN RULES (NEW IN v1.1) ===
PCB DESIGN RULES:
Stackup (4 layers): F.Cu signal / In1 GND / In2 PWR / B.Cu signal.

Controlled impedance:
  - USB (including FT231XS D+/D-):  90 ohm differential
  - GPDI / HDMI pairs:              100 ohm differential, length matched in-pair
  - GPIO +/- pairs:   matched length, NOT controlled impedance when routed
                      on In2 (see layer rules)
  - SDRAM:                          50 ohm single-ended target (not critical at
                                    100 MHz), length matched
  - Camera (OV7670, parallel DVP), SD card, ADC, audio, LEDs, UART:
                                    no impedance control, default net class
Trace widths and gaps: see Net Classes in the PCB Board Setup.

Layer rules:
  - High-speed nets (SDRAM, GPDI, USB) stay on F.Cu / B.Cu.
  - Do not cut In1 (GND) under high-speed traces.
  - Ten low-speed GPIO pins are routed on In2: GPIO21+/-, 22A/B, 24A/B,
    26A/B, 27A/B (longest run 48 mm). Use slow slew rate and low drive
    in the constraints. Do not use these pins for fast signals. GPIO21+/-
    is not suitable as a high-speed or LVDS pair.
  - ESD / TVS diodes sit at the connector with no stub.
  - Power traces are sized by current, pours for the 1.1V core.
```

```
=== REFERENCE DOCUMENTS USED THROUGHOUT ===
REFERENCE DOCUMENTS USED THROUGHOUT:
FPGA CORE
- ECP5 and ECP5-5G Family Datasheet — FPGA-DS-02012
- ECP5 sysI/O Usage Guide — FPGA-TN-02032
- ECP5 Power Decoupling and Bypass Filtering App Note —
FPGA-TN-02115
- ECP5 sysCONFIG Usage Guide (TN1260) — boot mode strapping,
pull-up
  requirements, SPI flash programming
- ECP5 Programming and Configuration Usage Guide — FPGA-TN-02050
- ECP5 Package Diagrams (mechanical/ball pitch/diameter) —
FPGA-DS-02053
- ECP5 Pin Migration (caBGA381) — pin-level ball/bank/differential
data
- Lattice Answer Database #738 — unused I/O pin handling

PER-COMPONENT DATASHEETS
- TLV62569 (switching regulator) — design equations, Table 4 (L/C
matrix)
- MAX11120-MAX11128 (ADC family) — MAX11125 specifics, SWCNV
mode
- IS25LP128 (SPI flash)
- IS42S16160J / MT48LC16M16A2 (SDRAM)
- FT231XS (USB-to-UART bridge) — pin functions, CBUS defaults
- PCA9306D (I2C level shifter) — DDC/EDID voltage translation
- BZT52C3V6 (zener diode) — HPD clamp
- OV7670 (camera sensor) — XCLK requirement, DVP timing
- CUI SJ-43516-SMT-TR (audio/video jack) — pinout

STANDARDS
- RS-170A / SMPTE 170M (composite video) — 75 ohm load impedance
origin
- USB Type-C spec — CC1/CC2 pulldown convention for legacy device
mode


=== CHANGES SINCE v1.0 ===
CHANGES SINCE v1.0:
- Bank 7 VCCIO documented as selectable 2.5V / 3.3V (was listed as
3.3V).
- Bank 8 VCCIO documented as 3.3V (was blank).
- Bank 6 description corrected (USB-UART, not JTAG).
- Added PCB design rules box (stackup, impedance targets, layer rules).
- Added note on true LVDS requirements for GPIO +/- pairs.
```

M3 M3 M3 M3

Power — File: Power_FPGAboard.kicad_sch

Flash — File: Flash_FPGAboard.kicad_sch

USB — File: USB_FPGAboard.kicad_sch

SDRAM — File: SDRAM_FPGAboard.kicad_sch

Analog — File: Analog_FPGAboard.kicad_sch

GPDI — File: GPDI_FPGAboard.kicad_sch

GPIO — File: GPIO_FPGAboard.kicad_sch

Camera — File: Camera_FPGAboard.kicad_sch

Status — File: Status_FPGAboard.kicad_sch

Sheet: /
File: FPGA v1.0.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 1/10

Switching Regulator + Decoup Caps

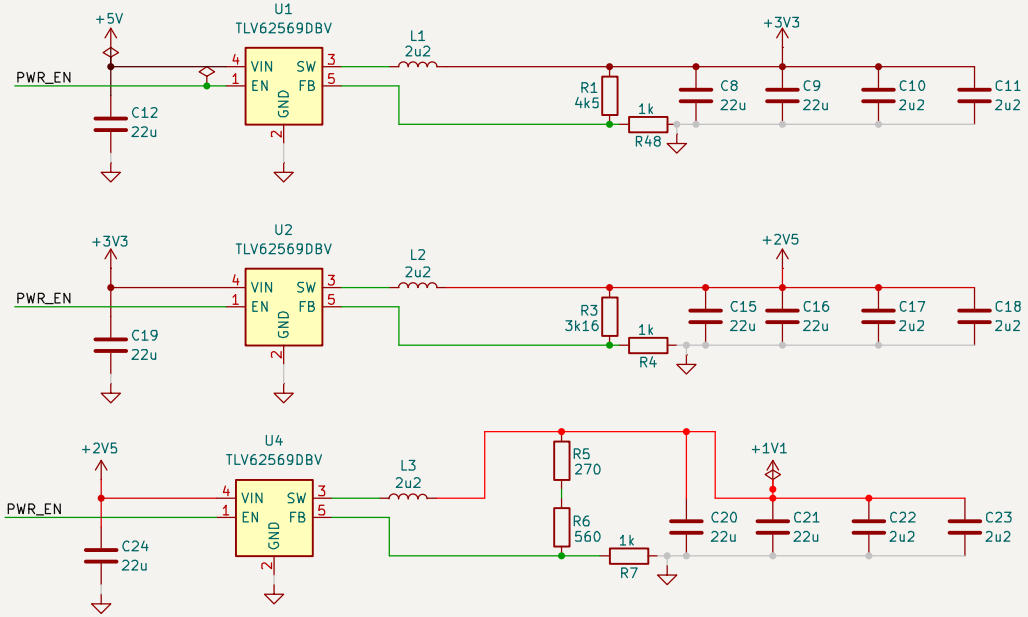
Notes:
8.2.2.2 Setting Output Voltage
Vout = V_FB * (1+ R1/R2)

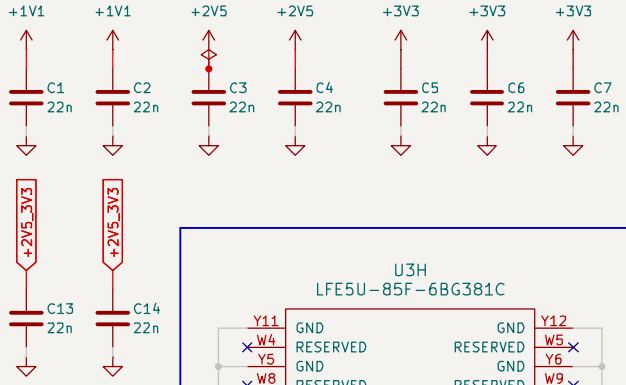
Relstor values found using:
R1/R2 = (Vout/V_FB)- 1

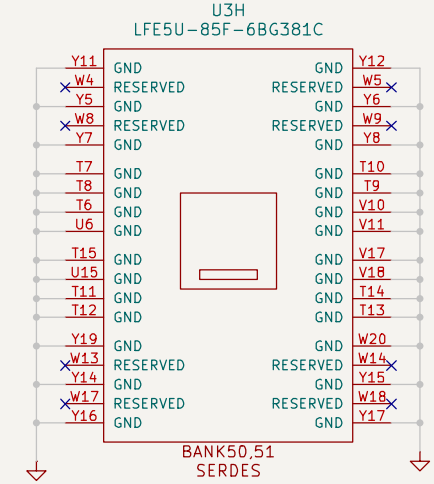
Capacitors + Inductor: see Table 4. Matrix of Output Capacitor and Inductor Combinations on
8.2.2.3 Output Filter Design of the Datasheet.

Power_EN: see 6.5 Electrical Characteristics - Logic Interface
OFF at 0.4V
ON at 1.2V

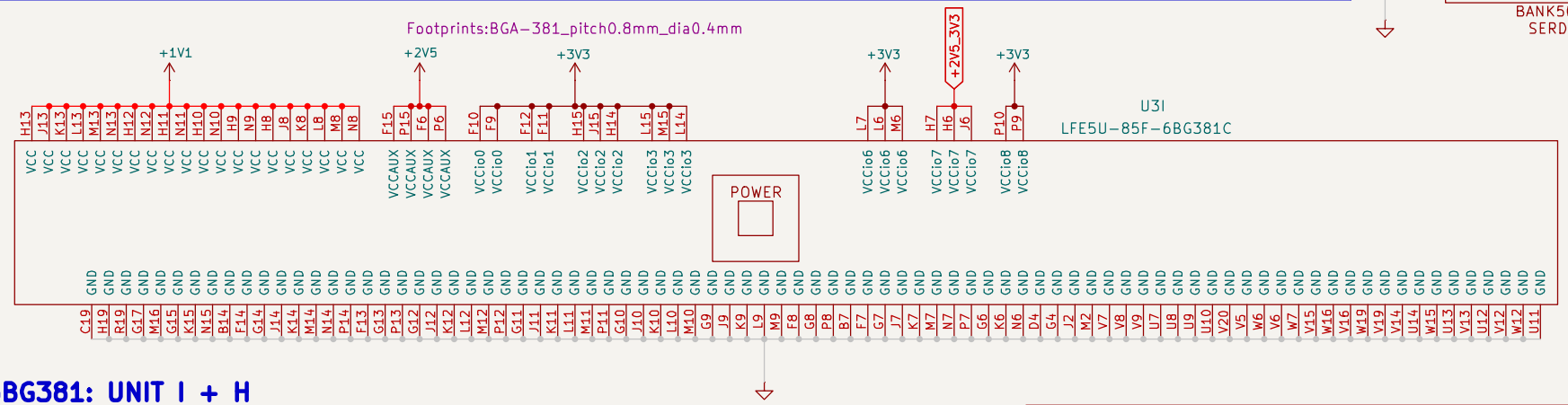
Decoupling Capacitors: see Section 3 of technical note FPGA-TN-02115-1.1







Footprints:BGA-381_pitch0.8mm_dia0.4mm



85F-6BG381: UNIT I + H

Notes:
VCC + VVCAUX are requirements set by Table 3.2 Reccomended Operating Conditions on the Datasheet.

LFE5U (non-UM) has no SERDES
RESERVED pins left NC. Confirm once GPDI bank is finalized.

Sheet: /Power/
File: Power_FPGAboard.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 9.0.2 | Id: 2/10

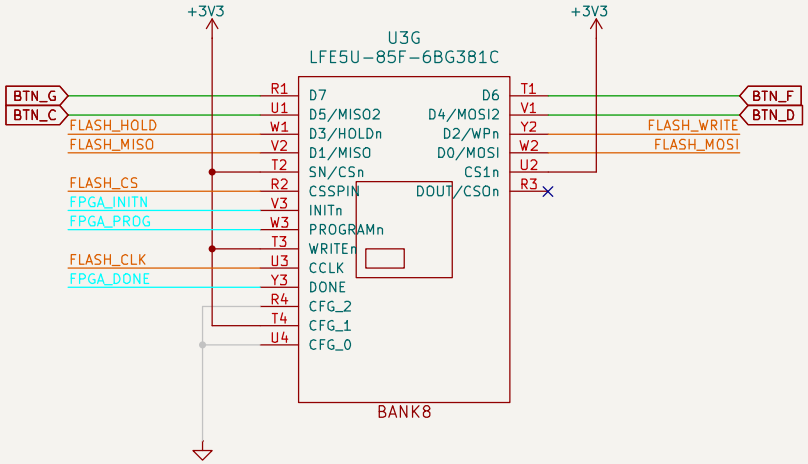
85F-6BG381: UNIT G

Notes:
Pull-up resistors: see TN1260 SysConfig Guide sec. 6.1 Master SPI modes
CGF_0, CGF_1, and CGF_2: Master sys config, see Figure 6.1

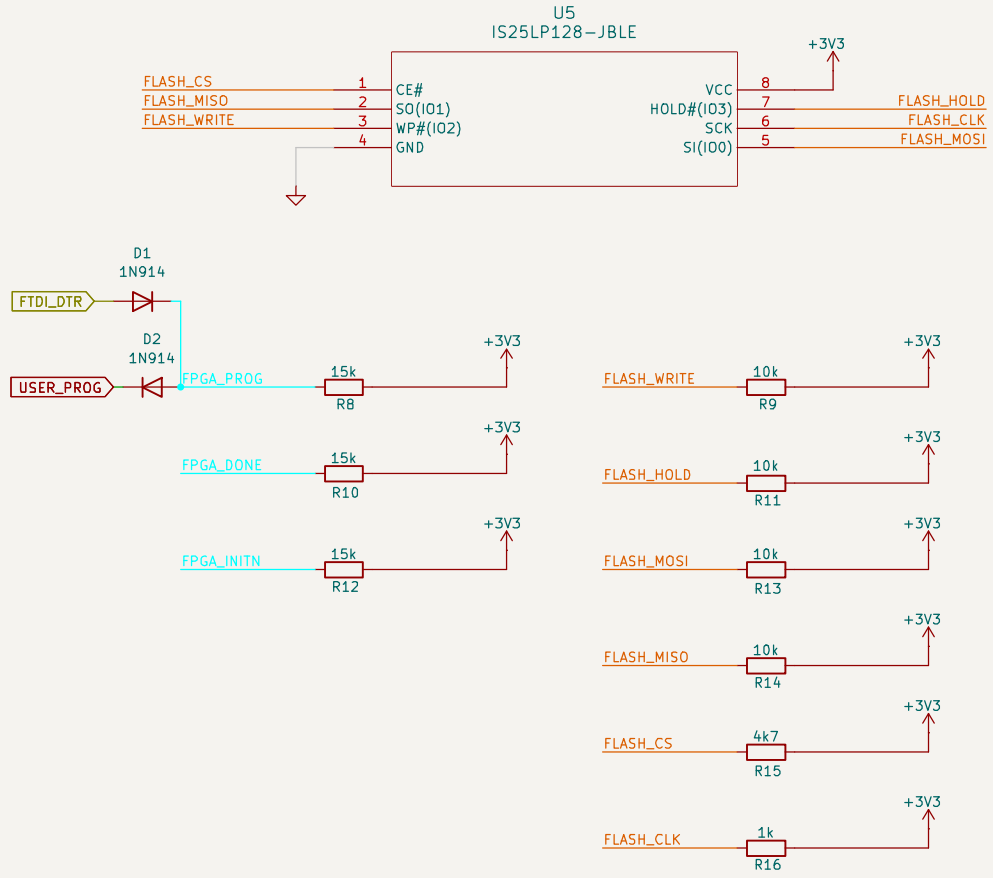
For programming Flash thru JTAG see Lattice FPGA-TN-02050.

USER_PROG: FPGA GPIO (Bank6) driven low by firmware to trigger self-reconfiguration/reset - not an external signal.

D9/D10 form a diode-OR network: either USER_PROG (firmware-triggered) or FTDI_DTR (USB-tool-triggered) can independently pull FPGA_PROG low to force reconfiguration. Diode orientation: cathode toward the trigger signal, anode toward the R14/3.3V node - current must flow from the pulled-up node into the low-driven source. Verify with multimeter: node should read ~0.6-0.7V when either trigger is asserted low.

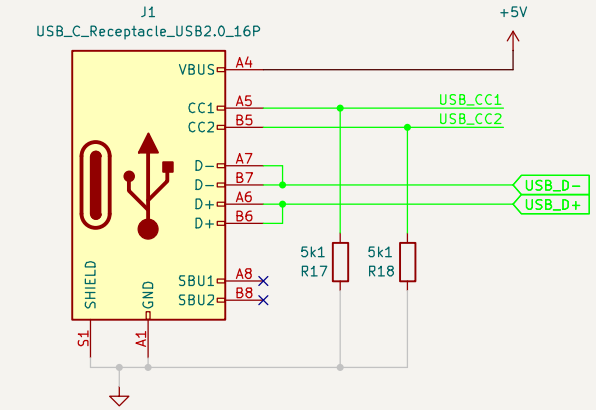


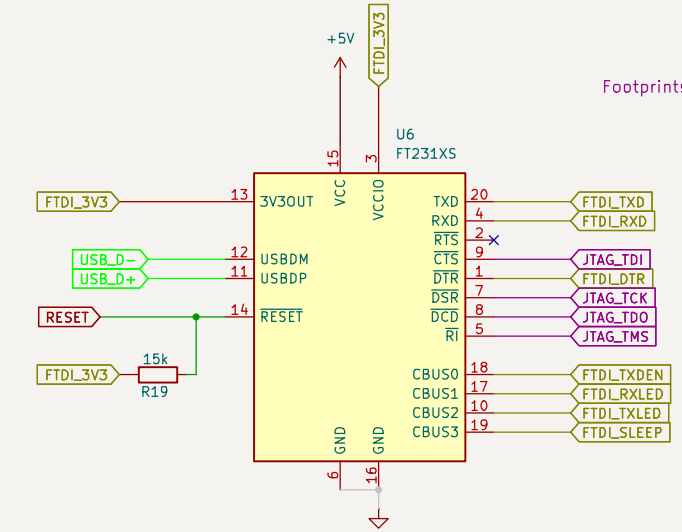
Programming Connector + Pull-up Resistors

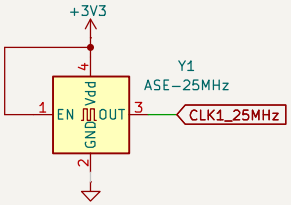


Sheet: /Flash/		
File: Flash_FPGAboard.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 3/10

USB-C + FTDI





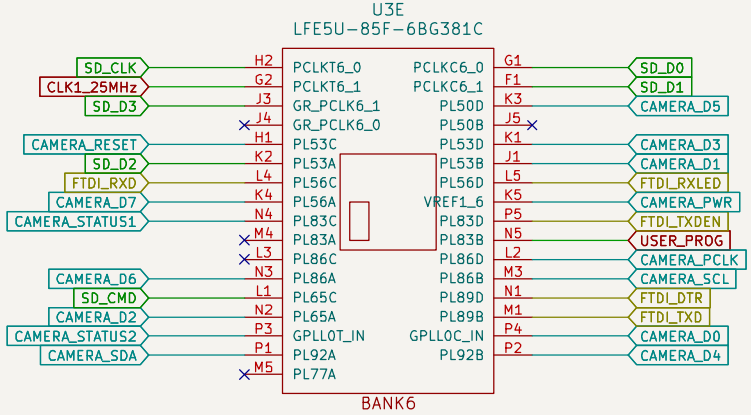


85F-6BG381: UNIT E + JTAG

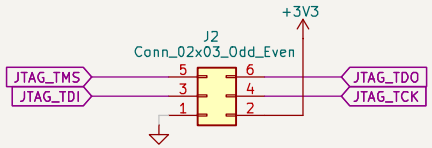
Notes:

FTDI_TXD/FTDI_RXD are named from the FT231XS's, not the FPGA's. FTDI_TXD is an FPGA INPUT (receive), FTDI_RXD is an FPGA OUTPUT (transmit). Verilog UART module: assign RX to FTDI_TXD, TX to FTDI_RXD.

Footprints:BGA-381_pitch0.8mm_dia0.4mm







Sheet: /USB/
File: USB_FPGAboard.kicad_sch

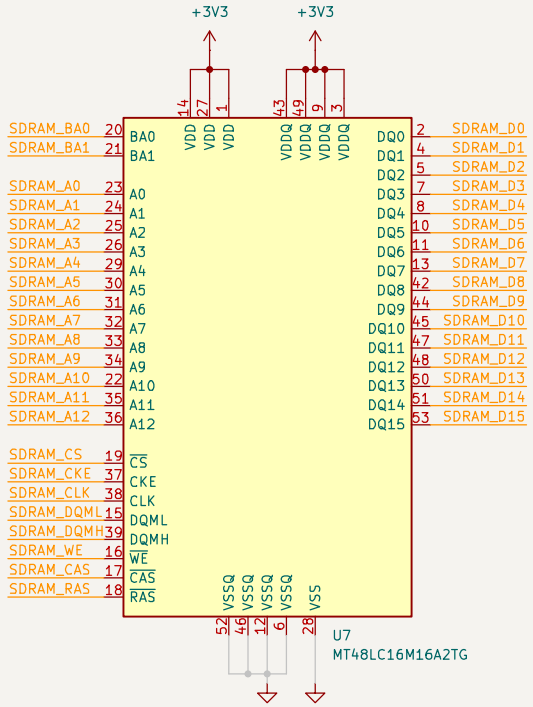
Title:

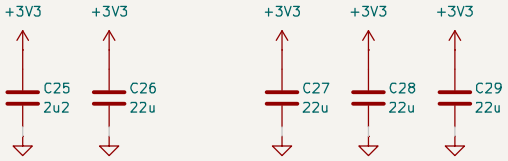
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 4/10

Footprints:BGA-381_pitch0.8mm_dia0.4mm

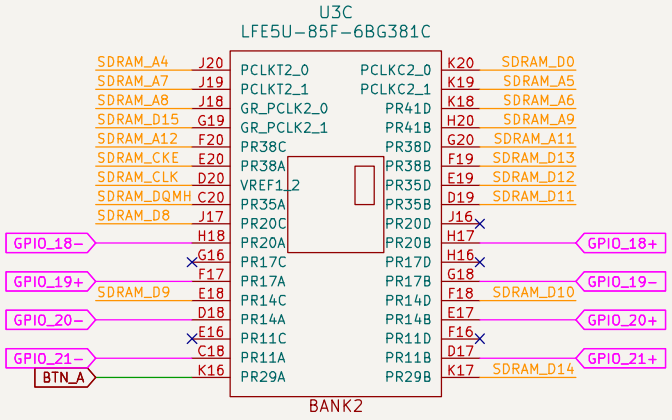
SDRAM – 256mb

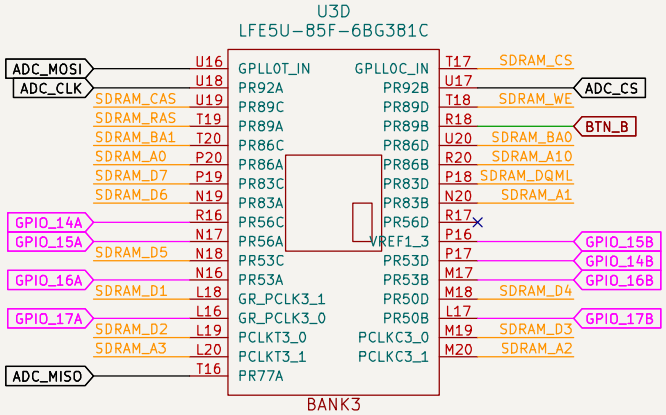
Notes: Refer to PIN ASSIGNMENT METHODOLOGY note on the tittle page for pin assignment info.





85F-6BG381: UNIT C + D





Sheet: /SDRAM/
File: SDRAM_FPGAboard.kicad_sch

Title:

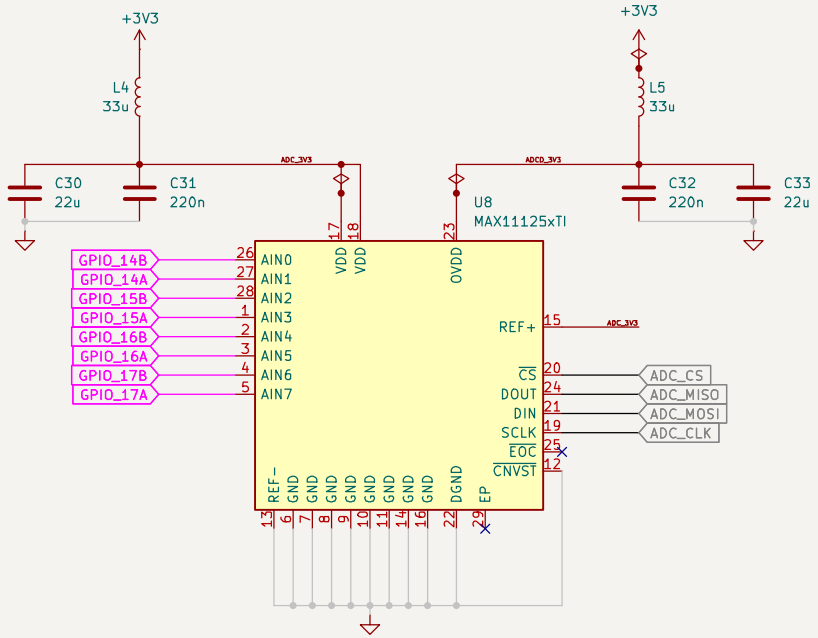
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 5/10

Analog-to-Digital Converter

Notes:
EP - Exposed Pad, on QFN/TQFN packages tied to GND for thermal dissipation
+ analog GND ref.

REF+ per Pin Description table, REF- is the differential reference's negative input, tied to GND here to run the ADC single-ended, rather than using the optional differential-reference mode. Routed through a separate filtered ADC3V3 rail (ferrite bead + local decoupling) rather than the main 3.3V rail, this isolation is not a datasheet requirement, it's standard analog-reference practice (same technique as Lattice's "Power Decoupling and Bypass Filtering" app note, Fig. 3.1), since REF+ directly sets the ADC's full-scale voltage and is noise-sensitive.

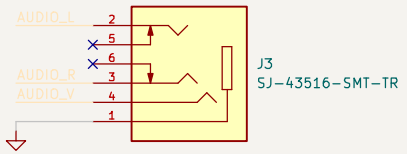
Selects SWCNV (software convert) mode - conversions are triggered entirely via an SPI register. See Internal Conversions with SWCNV in the Datasheet

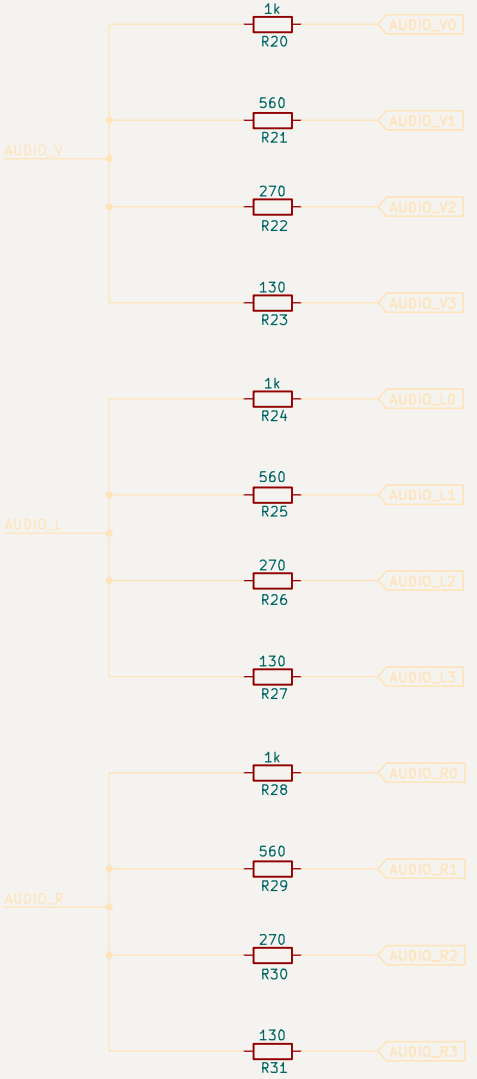


Audio Jack

Resistor Values found using:
R_parallel = 1/(1/R1+1/R2+1/R3+1/R4)
R_parallel = 70.51 ohm - 75ohm

75ohm target = composite video standard load impedance (RS-170A / SMPTE 170M: 1Vpp video, unbalanced, terminated 75ohm). This values minimizes signal loss for this application





Sheet: /Analog/
File: Analog_FPGAboard.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 6/10

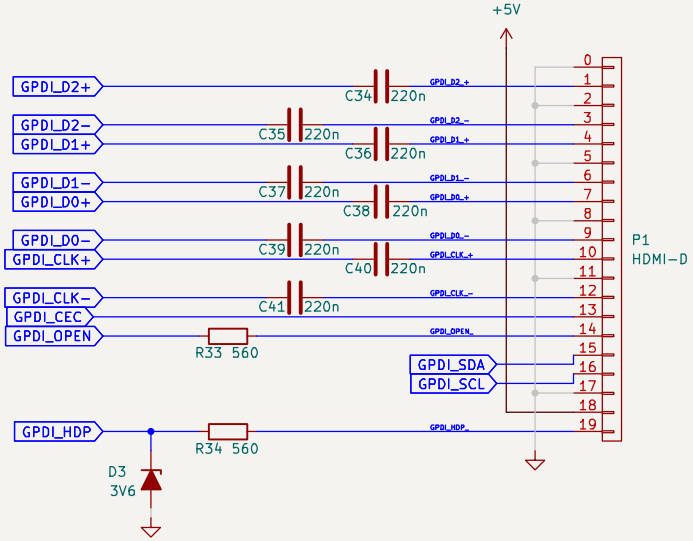
Cap Connector + Voltage Translator

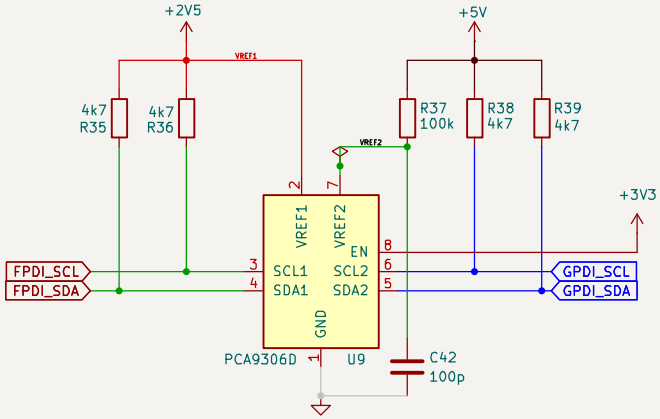
Notes:

GPDI_HPD (Hot Plug Detect): tells firmware whether a display is actually connected before attempting to output video. Real signal, worth using in firmware logic. It's connected to 3.6V via zener (D13) since the connector side runs at 5V and the FPGA GPIO doesn't tolerate that.

GPDI_OPEN: spare pin, no defined function in this design. Left pulled up, unused for now.

GPDI_CEC: present with pull-up but not implemented as a feature, CEC (remote-control-over-HDMI) is out of scope for this board.



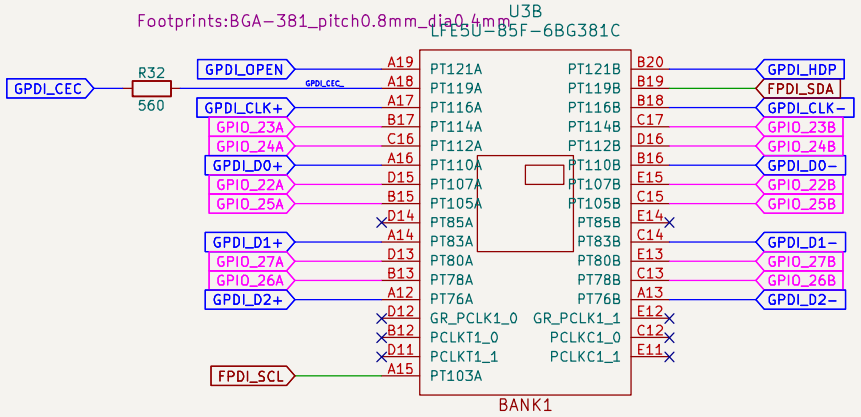


85F-6BG381: UNIT B

Notes:

DDC/EDID (GPDI_SCL/SDA + PCA9306D level shifter): lets firmware read a connected display's supported resolutions instead of blindly outputting one fixed format. The level shifter exists because DDC runs at 5V on the connector side vs. our FPGA's lower logic voltage required for chip safety regardless of whether EDID is ever read. ACTUAL EDID PARSING IS NOT YET IMPLEMENTED IN FIRMWARE, this is a "nice to have, add later" feature. Fine to hardcode one output resolution for now and revisit this once there's bandwidth.

Reprogram-trigger diode network: see Flash sheet for how USER_PROG and FTDI_DTR both independently pull FPGA_PROG low.



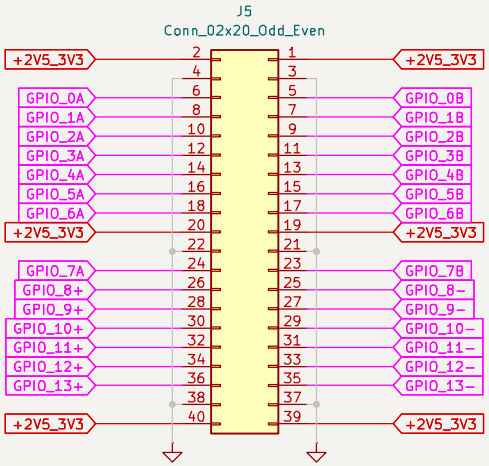
Sheet: /GPDI/
File: GPDI_FPGAboard.kicad_sch

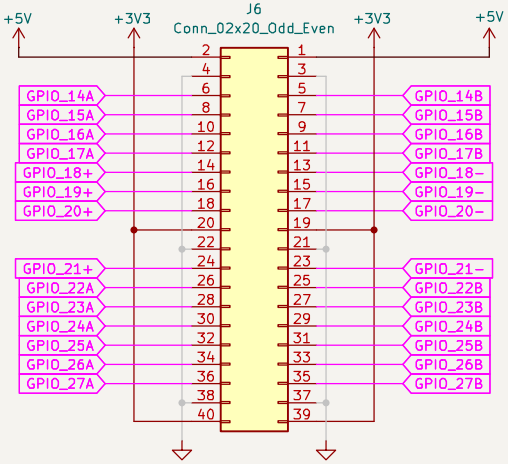
Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 9.0.2 | Id: 7/10

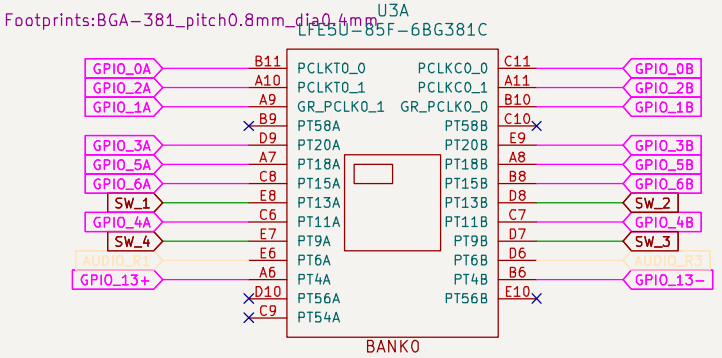
GPIO headers

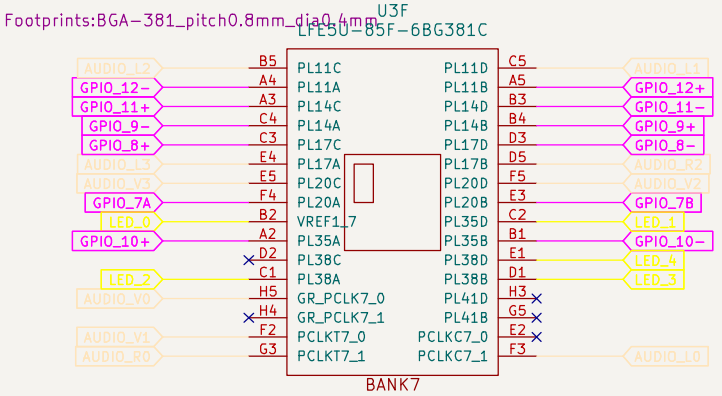
Notes: Refer to PIN ASSIGNMENT METHODOLOGY note on the tittle page for pin assignment info.





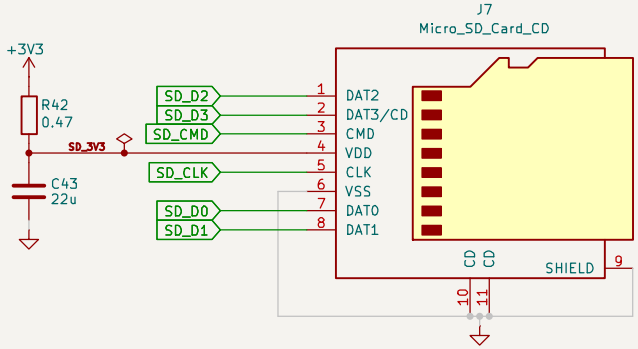
85F-6BG381: Unit A + F



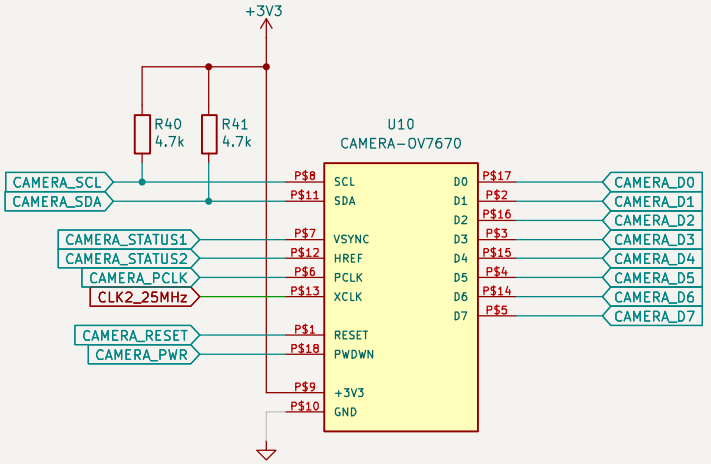


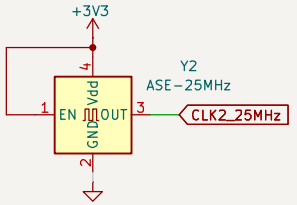
Sheet: /GPIO/	
File: GPIO_FPGAboard.kicad_sch	
Title:	
Size: A4	Date:
KiCad E.D.A. 9.0.2	Id: 8/10

SD Card



Camera

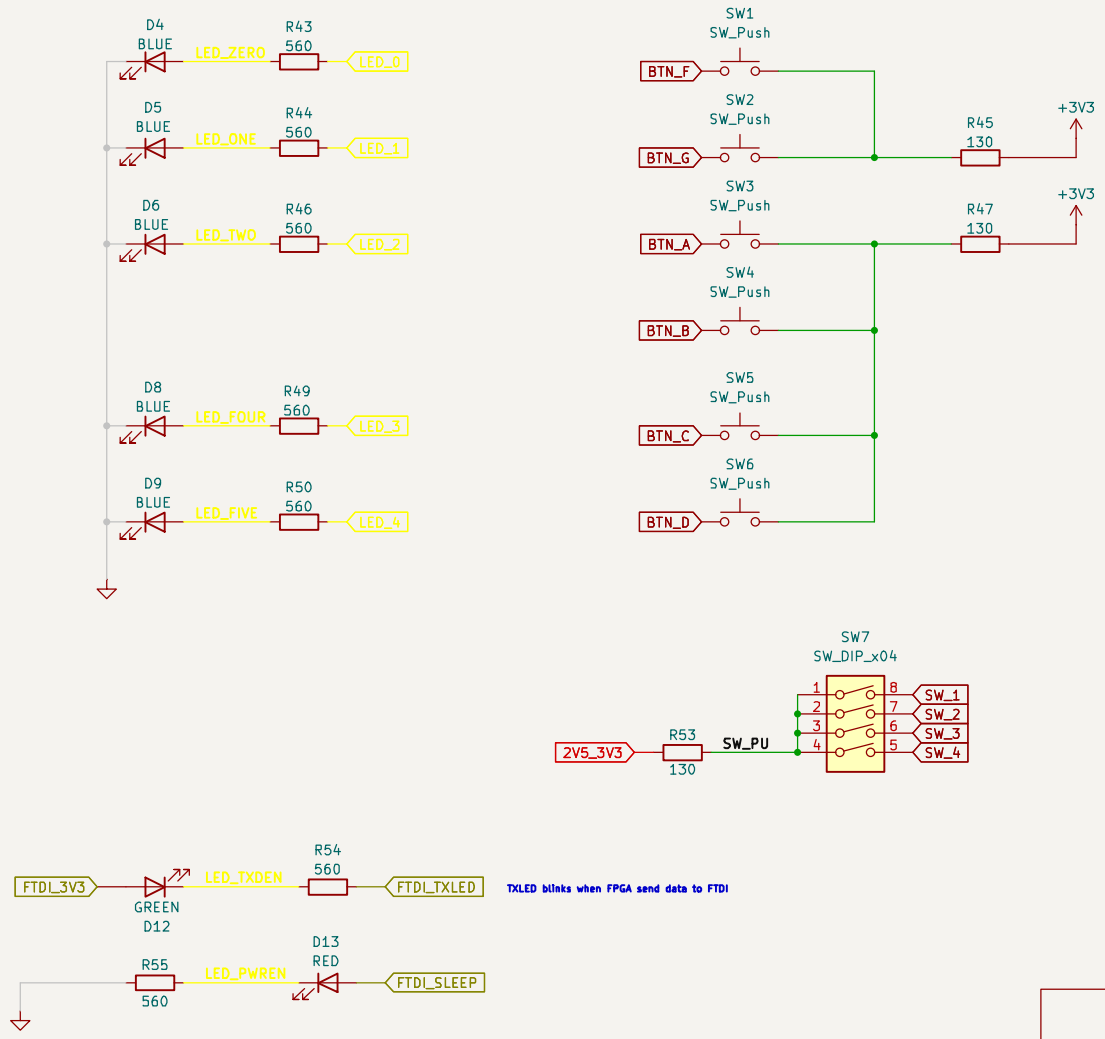




Sheet: /Camera/
File: Camera_FPGAboard.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 9/10



Sheet: /Status/		
File: Status_FPGAboard.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 10/10